

CityBES Dataset Explained

The City Building Energy Saver (CityBES) is a web-based tool to support urban-scale building energy modeling. Users can create their own dataset and upload it to CityBES for visualization, modeling, and analysis. This document describes the requirements of datasets accepted by CityBES.



Figure 1. Screenshot of visualization of a CityBES dataset

Basic Requirements

Required parameters

CityBES prefers dataset in [GeoJSON](#) format, which is a popular geospatial data interchange format based on JavaScript Object Notation (JSON). In the GeoJSON dataset, CityBES requires the **geometry** of each building, together with the following building properties:

- **Year of built** (integer)
- **Building height** (float): in meter
- **Number of stories** (integer)

- **Building type** (string): Please specify the building use type in the following words (case-insensitive) recognizable by CityBES. Currently the building types that can be modeled by CityBES include:
 - **Office** (in United States, Italy, and Chile): office, 211, 212, 500, bz, o, oa, oah, oal, oam, oat, obh, obl, obm, oc, och, ocl, ocm, omd, oz
 - **Retail** (in United States): retail, 202, 206, 208, 209, c, c1, cz, rh1
 - **Restaurant** (in United States): restaurant, 201
 - **Hotel** (in United States): hotel, h, h1, h2, hc, 110
 - **Single Family House** (in United States California and Oregon): 101, d, cos, dbm, th, thbm, single family, single-family, residential-single family, single-family building, single family building, single-family house, single family house
 - **Multi Family House** (in United States California and Oregon, and Italy): 102, 104, 103, z, ZBM, ZEU, da, da15, da5, a, a15, a5, co, f, f15, f5, ti15, tia, tic, tic5, tif, multi family, multi-family, residential-multi family, multi-family building, multi family building, multi-family house, multi family house
-

Optional properties:

- **Unique building id** (string): This is optional but recommended. Users can use it to map the simulation results to each building in the dataset. The id for each building should not be duplicated within your dataset.
 - **Building name** (string): Will be shown on the visualization page when the building is selected. Can be building address or other names
-

The building **geometry** required by CityBES now is only the building footprint. The coordinates in the building **geometry** should be in the local [Coordinate Reference Systems \(CRS\)](#) specified at the beginning of the dataset in [EPSG](#) code, which is a unique numerical identifier for a CRS and other geodetic properties.

Note that the current GeoJSON [standard](#) requires CRS84 (the World Geodetic System 1984 (WGS 84) [WGS84] datum, with longitude and latitude units of decimal degrees), and aborted the usage of alternative CRS, but CityBES currently still requires a local CRS to your dataset for higher accuracy of calculating area, distance, etc. This might be changed to align with the GeoJSON standard in the future.

Please see accepted CRS below. CRSs for other districts can be added per request.

- [2227](#): United States - California north of 36.5°N

- [32118](#): United States - New York
- [2992](#): United States - Oregon and Washington
- [2229](#): United States - California south of 36.5°N
- [2253](#): United States - Michigan
- [3003](#): Italy zone 1
- [3004](#): Italy zone 2
- [6707](#): Italy - onshore and offshore
- [3763](#): Portugal
- [8838](#): Saudi Arabia - onshore and offshore
- [25830](#): Europe - onshore and offshore
- [31982](#): Latin America
- [28600](#): Qatar - onshore
- [32718](#): Chile
- [32161](#): United States - Puerto Rico and US Virgin Islands - onshore

! CityBES will validate the building properties when processing the uploaded dataset, and exclude those buildings if

- any of the required building properties is missing or invalid
- gross floor area (calculated from geometry) < 30m², or
- floor-to-floor height ([Building height](#) / [Number of stories](#)) not between 2.5m and 6m

Example dataset

Here's an example dataset with one building. Note that:

- The required building properties in the dataset can be named as the user likes. The user needs to specify the field name for each property when they upload the dataset to CityBES (see [Figure 2](#) as an example)
- The CRS is specified as EPSG 2229, and the coordinates are all projected into this CRS
- The unit of building height needs to be meter
- To add more buildings, just add more elements with the same structure under the [features](#) array

```
{
  "type": "FeatureCollection",
  "name": "EXAMPLE CAMPUS",
  "crs":
```

```
{
  "type": "name",
  "properties": {
    "name": "urn:ogc:def:crs:EPSG::2229"
  }
},
"features": [
  {
    "type": "Feature",
    "properties": {
      "BLD_ID": "bldg-1",
      "Height_m": 10.4,
      "NoStorey": 3,
      "UseType": "multi family house",
      "YrBltd": 1950,
      "BldgName": "residential bldg 1"
    },
    "geometry": {
      "type": "MultiPolygon",
      "coordinates": [
        [
          [1726676.766, 5032463.609],
          [1726678.9250325235, 5032463.305179942],
          [1726677.9997956986, 5032455.971709897],
          [1726677.3868230395, 5032450.396525458],
          [1726675.42252002, 5032435.41347515],
          [1726674.5751726916, 5032429.393067757],
          [1726673.8725071994, 5032422.831913434],
          [1726671.4860534582, 5032423.042317734],
          [1726665.3901083919, 5032423.696244818],
          [1726666.313, 5032432.148],
          [1726665.688, 5032432.172],
          [1726666.0, 5032435.547],
          [1726666.797, 5032435.453],
          [1726669.0, 5032452.273],
          [1726668.203, 5032452.383],
          [1726668.563, 5032455.773],
          [1726669.266, 5032455.594],
          [1726670.422, 5032464.398],
          [1726676.766, 5032463.609]
        ]
      ]
    }
  ]
}
```

```
]
}
}
]
}
```

Prototype District 608

Unique index (required)	20220315
City (required)	Los Angeles
Unique building dataset name (required)	LA - Central City (local)
Data file type (select one)	GeoJSON
Building type filed (required)	UseType
Measured height filed (required) (unit: m)	Height_m
Year built filed (required)	YrBlt
Number of stories filed (required)	NoStorey
Data model file (required)	Choose File No file chosen
Current uploaded data file	La-central city-trim-new.geojson
User-defined building id filed (optional)	BLD_ID
Building name filed (optional)	BldgName

Figure 2. Filled form when uploading the example dataset

The required building information can come from different data sources. [This document](#) demonstrates a workflow of merging different data sources to create a CityBES dataset, using GIS tools such as QGIS.

For Advanced Users

Apart from the basic required building properties, CityBES also allows users to customize some of other building properties through the `additional_json` field. These additional properties usually affect the building energy performance significantly, so it is useful when the user has

extra information for each building and wants to have more accurate simulation results. If the desired building type is not currently available in the district the user wants to model, this can also be an alternative approach to customize the buildings based on existing building prototypes.

The GeoJSON below show the the same example building with the `additional_json` field.

```
{
  "type": "FeatureCollection",
  "name": "EXAMPLE CAMPUS",
  "crs": {
    "type": "name",
    "properties": {
      "name": "urn:ogc:def:crs:EPSG::2229"
    }
  },
  "features": [
    {
      "type": "Feature",
      "properties": {
        "BLD_ID": "bldg-1",
        "Height_m": 10.4,
        "NoStorey": 3,
        "UseType": "multi family house",
        "YrBltn": 1950,
        "BldgName": "residential bldg 1",
        "additional_json": {
          "lighting_power_density": 15,
          "equipment_power_density": 10,
          "occupancy_density": 0.1,
          "outdoor_air_flow_per_person": 0.01,
          "roof_insulation_r_value_ip": 18.90033831,
          "roof_solar_reflectance": 0.4,
          "wall_insulation_r_value_ip": 7.44247035,
          "window_u_value_ip": 0.582010582,
          "window_shgc": 0.47,
          "window_visual_transmittance": 0.85,
          "window_wall_ratio": 0.3,
          "shw_system_type_id": 1,
          "shw_thermal_efficiency": 0.95,
          "hvac_system_type_id": 6,

```

```

    "hvac_cooling_cop": 3.54,
    "hvac_heating_efficiency": 0.9,
    "hvac_fan_efficiency": 0.56,
    "heating_setpoint": 68,
    "cooling_setpoint": 72,
    "schedule_set":
    {
        "heating_setpoint_schedule":
        {
            "1/1-4/30":
            {
                "weekday":
[18,18,18,18,18,18,18,18,18,18,18,18,18,18,18,18,18,18,18,18,18,18],
                "weekend":
[19,19,19,19,19,19,19,20,20,20,20,20,20,20,20,20,20,20,20,20,20,19]
            },
            "5/1-12/31":
            {
                "weekday":
[5,5,5,5,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24],
                "weekend":
[4,4,4,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]
            },
        },
        "cooling_setpoint_schedule":
        {
            "1/1-5/31":
            {
                "weekday":
[28,28,28,28,28,28,28,28,28,28,28,28,28,28,28,28,28,28,28,28,28,28],
                "weekend":
[28,28,28,28,28,28,28,28,28,28,28,28,28,28,28,28,28,28,28,28,28,28]
            },
            "6/1-12/31":
            {
                "weekday":
[25,25,25,25,25,25,23,23,23,23,23,23,23,23,23,23,23,25,25,25,25,25,25],
                "weekend":
[25,25,25,25,25,25,23,23,23,23,23,23,23,23,23,23,23,25,25,25,25,25,25],
                "summer_design_day":
[23,23,23,23,23,23,23,23,23,23,23,23,23,23,23,23,23,23,23,23,23,23,23]
            }
        }
    },
    },
    },

```

```

"geometry":
{
  "type": "MultiPolygon",
  "coordinates": [
    [
      [
        [1726676.766, 5032463.609],
        [1726678.9250325235, 5032463.305179942],
        [1726677.9997956986, 5032455.971709897],
        [1726677.3868230395, 5032450.396525458],
        [1726675.42252002, 5032435.41347515],
        [1726674.5751726916, 5032429.393067757],
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        [1726666.313, 5032432.148],
        [1726665.688, 5032432.172],
        [1726666.0, 5032435.547],
        [1726666.797, 5032435.453],
        [1726669.0, 5032452.273],
        [1726668.203, 5032452.383],
        [1726668.563, 5032455.773],
        [1726669.266, 5032455.594],
        [1726670.422, 5032464.398],
        [1726676.766, 5032463.609]
      ]
    ]
  ]
}

```

Here's a full list of available parameters that can be customized through `additional_json`, as well as their explanation and accepted options. (Since CityBES uses EnergyPlus™ as the simulation engine, many of these parameters are directly used in EnergyPlus™ so the definitions are similar)

- `lighting_power_density` (float): (average) electrical load of interior lighting for the building in **W/m²**
- `equipment_power_density` (float): (average) electrical load of interior equipment (such as home appliances and electric) for the building in **W/m²**
- `occupancy_density` (float): in **person/m²**

- **outdoor_air_flow_per_person** (float): for HVAC system, in $\text{m}^3/(\text{s}\cdot\text{person})$,
- **infiltration_flow_rate_per_exterior_wall_area** (float): in $\text{m}^3/(\text{s}\cdot\text{m}^2)$,
- **roof_insulation_r_value_ip** (float): R-value of the insulation layer of the roof construction, in $\text{ft}^2\cdot\text{°F}\cdot\text{h/Btu}$; if you have the value in SI unit ($\text{K}\cdot\text{m}^2/\text{W}$), convert it by *5.68
- **roof_solar_reflectance** (float): **0 < value < 1**
- **wall_insulation_r_value_ip** (float): R-value of the insulation layer of the exterior wall construction, in $\text{ft}^2\cdot\text{°F}\cdot\text{h/Btu}$; if you have the value in SI unit ($\text{K}\cdot\text{m}^2/\text{W}$), convert it by *5.68
- **window_u_value_ip** (float): U-value of the window, in $\text{Btu}/(\text{h}\cdot\text{ft}^2\cdot\text{°F})$
- **window_shgc** (float): Solar Heat Gain Coefficient of the window, **0 < value < 1**
- **window_visual_transmittance** (float) VT, **0 < value < 1**
- **shw_system_type_id** (integer): service hot water system type, choose from
 - 0: Gas storage water heater
 - 1: Gas tankless water heater
 - 2: Electric storage water heater
 - 3: Electric tankless water heater
 - 4: Heat pump water heater
- **shw_thermal_efficiency** (float): rated service hot water system efficiency
 - if shw_system_type_id is not specified, or shw_system_type_id is not 4: **0 < value < 1**
 - else: **value > 1**
- **hvac_system_type_id** (integer): choose from
 - 0: Packaged single zone rooftop air conditioner
 - 1: Packaged rooftop VAV with reheat for all stories
 - 2: Packaged rooftop VAV with reheat for each story
 - 3: Packaged single zone rooftop heat pump
 - 4: Packaged rooftop VAV with pfp for all stories
 - 5: Packaged rooftop VAV with pfp for each story
 - 6: Packaged rooftop VAV with gas furnace for all stories
 - 7: Packaged rooftop VAV with gas furnace for each story
 - 8: Packaged single zone rooftop air conditioner VAV
 - 9: Packaged single zone rooftop heat pump VAV
 - 10: Packaged rooftop VAV heat pump for all stories
 - 11: Packaged rooftop VAV heat pump for each story
 - 12: Variable refrigerant flow for all stories
 - 13: Variable refrigerant flow for each story
 - 14: VAV with reheat: center plant
 - 15: DOAS with CAV: center plant

- 16: DOAS with VAV: center plant
- 17: CAV with reheat: center plant
- 18: Packaged terminal heat pump
- 19: Packaged CAV rooftop heat pump for all stories
- 20: Split system air conditioner with gas furnace for all stories, single thermostat
- 21: Heating with radiator and PTAC for cooling for single zone
- 22: Heating with radiator and no cooling for single zone
- 23: Window AC with wall heater
- 24: Portable AC with wall heater
- 25: Mini split heat pump
- 26: Split system air conditioner with gas furnace for single zone
- 27: Swamp cooler: for all stories with wall heater: for single zone
- 28: Swamp cooler with wall heater for single zone
- 29: Split system air conditioner with gas furnace for all stories - vav heat cool
- 30: Centralized heating with radiator and no cooling for all stories
- 31: Centralized heating with radiator and split air conditioner for cooling for all stories
- 32: Heating with radiator and split air conditioner for cooling for single zone
- **hvac_cooling_cop** (float): rated COP of HVAC system, only required when hvac_system_type_id is specified
- **hvac_heating_efficiency** (float): rated heating efficiency of HVAC system, only required when hvac_system_type_id is specified
- **hvac_fan_efficiency** (float): rated fan efficiency, only required when hvac_system_type_id is specified; **0 < value < 1**
- **heating_setpoint** (float): constant thermostat heating setpoint, in °F
- **cooling_setpoint** (float): constant thermostat cooling setpoint, in °F
- **schedule_set** (object): use key-value pairs to customize the schedules in detail
 - schedules that can be customized are:
 - **heating_setpoint_schedule** : thermostat heating setpoint, in °C
 - **cooling_setpoint_schedule** : thermostat cooling setpoint, in °C
 - **occupancy_schedule** : occupancy in fraction, **0 ≤ value ≤ 1**
 - **interior_lighting_schedule** : fractions to modify the design level energy of interior lights; the electrical input for lighting in a particular timestep is the product of the design level and the value of this schedule in that timestep
 - **electric_equipment_schedule** : fractions to modify the design level energy of interior electric equipment; the electrical input for interior equipment in a particular timestep is the product of the design level and the value of this schedule in that timestep

- **gas_equipment_schedule** : fractions to modify the design level energy of gas equipment; the energy input for gas equipment in a particular timestep is the product of the design level and the value of this schedule in that timestep
- **hvac_schedule** : HVAC availability schedule, 1 for the times the system is available and 0 for not available
- **occupancy_activity_schedule** : the schedule for the amount of heat gain per person in the zone under design conditions, in **W/person**; activity level for most office activities can range from 100 to 150 Watts per person; for other activities, please refer to ASHRAE Standard 55
- **zone_infiltration_schedule** : fractions relative to the maximum design volume flow rate of air infiltration, $0.0 \leq \text{fraction} \leq 1.0$
- **oa_control_schedule** : on-off schedule to control the outdoor air flow rate, 1 for on and 0 for off
- **water_use_schedule** : the flow rate fraction relative to the peak demanded total water flow rate, $0.0 \leq \text{fraction} \leq 1.0$
- each schedule can be specified for multiple date periods for the whole year; for each date period, one weekday schedule and one weekend schedule are allowed; for setpoint schedules, **summer_design_day** and **winter_design_day** schedules can be added in additional for sizing the equipment. See the format elaborated below:

```

"xxx_schedule": {
  "01/01-endDate": {
    "weekday": [value1, value2...value24],
    "weekend": []
  },
  "startDate2-endDate2": {
    "weekday": [],
    "weekend": []
  },
  "startDate3-endDate3": {
    "weekday": [],
    "weekend": []
  }
}

```